

Aaron Low Weng Soon

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SUMMARY

Senior robotics engineer with six years in autonomous vehicles, specialising in multi-sensor calibration and fusion. I've worked on systems that utilise a combination of deep learning and classical methods like factor graphs, multi-view geometry, and SLAM. I work end to end, developing the algorithms and building the scalable, automated tooling and infrastructure that makes them fast to iterate on and reliable to run.

WORK EXPERIENCE

Wayve

2024 - Present

Senior Robotics Engineer (Sensor Calibration and Fusion) • 2024

London, United Kingdom

- Designed and implemented multi-sensor calibration algorithms (camera, IMU, lidar) combining deep learning with multi-view geometry and SLAM.
- Built scalable automated pipelines running calibration continuously supporting a fleet of vehicles spread across multiple locations globally.
 - Our improvements led to a 10x speedup in turnaround time for calibrating a vehicle.
- Built an end-to-end deep learning pipeline for training and evaluating visual feature extraction and calibration models, covering dataset generation, curation, and loading.
- Implemented metrics and automated validation that detect miscalibration and drift, catching issues before they reached downstream users.
- Developed internal tooling to deploy, track, and maintain per-vehicle calibrations across the fleet.
- Led technical direction, mentored junior engineers, and helped shape hiring.

Motional

2020 - 2024

Senior Research Engineer (Sensor Calibration) • 2023 - 2024

Singapore

Research Engineer (Sensor Calibration) • 2020 - 2023

- Developed solutions for sensor (cameras, LIDARs, radars, IMU) calibration and validation of autonomous vehicles.
- Researched deep learning solutions for sensor calibration as well as implementing the infrastructure for training and evaluation.
- Built and maintained cloud-based systems for continuous validation of vehicle sensor data.
 - Our automated system ran daily on a fleet of hundreds of vehicles allowing data-driven calibrations.
- Automated dataset generation and curation for algorithm research.
- Worked on vehicle hardware to deploy online sensor calibration algorithms.

Hyundai-Aptiv Autonomous Driving Joint Venture

2020

Autonomous Vehicle Intern (Sensor Calibration)

Singapore

- Researched deep learning for camera-to-LiDAR calibration and deployed a **RegNet** based model into production with TensorRT.
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PUBLICATIONS

- **UniCal: a Single-Branch Transformer-Based Model for Camera-to-LiDAR Calibration and Validation** — a Transformer architecture using self-attention to perform camera-to-LiDAR calibration and validation in a single branch.
 - **Depth to Colour Translation for 3D Hand Pose Estimation From Monocular RGB With Generative Adversarial Networks** — MEng thesis, Imperial College London.
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EDUCATION

Imperial College London

2018 - 2019

PhD Researcher, Machine Learning and Computer Vision

- **Research area:** 3D Pose Estimation

Imperial College London

2014 - 2018

Electrical and Electronic Engineering MEng

- **First Class Honours** • **Dean's List (top 10% of class)** Year 2
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TEACHING

HELP University

2019 - 2020

Lecturer, Faculty of Computing and Digital Technology

- Lectured introductory programming and presented deep learning tutorials to both staff and students.
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PROJECTS

Autonomous Snack Delivery Android (ASDA)

2017

Robot Navigation

- Autonomous robot that manoeuvres a building and takes an elevator to obtain and deliver snacks.

ParkWare

2016

Machine Learning and Web

- **Prize Winner ("Best use of Amazon Web Services")** Parking space detection web service using machine learning to detect cars in parking lots.

Without Abandonware

2021

Unity/C# Developer

- **Featured by GitHub** in "Top entries from Game Off 2021". A 2D platformer written in Unity with C#.
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TECHNICAL SKILLS

Programming: Python • C++ • C#

ML and Robotics: PyTorch • TensorFlow • ROS • TensorRT • Multi-view Geometry • SLAM • GTSAM

Infrastructure: AWS • Azure • Terraform • Docker • Jenkins • Serverless • Bazel • CMake

Tools and Data: Git • Linux • Bash • Pandas • SQL • Unity

Languages: English (Native) • Malay (Limited Working)
